

Student 8 **H Harshavardhan** 192511171RD

4 / 10 submitted

Q1. Reverse a list

— Not Submitted —

Q2. Reverse a string

— Not Submitted —

Q3. Searching

CODE

```
#include<stdio.h>
int main()
{
    int n,key,low=0,found=0;
    printf("enter the size of array:\n");
    scanf("%d",&n);
    int arr[n],high=n-1;
    printf("enter %d sorted elements\n",n);
    for(int i=0;i<n;i++)
        scanf("%d",&arr[i]);
    printf("enter the element to search:\n");
    scanf("%d",&key);
    while(low<=high){
        int mid=(low+high)/2;
        if(arr[mid]==key){
            printf("element found at index %d\n",mid);
            found=1;
            break;
        }
        if(arr[mid]<key)
            low=mid+1;
        else
            high=mid-1;
    }
    if (!found)
        printf("element not found\n");
    return 0;
}
```

INPUT

```
6
10 20 30 40 50 60
40
```

OUTPUT

```
enter the size of array:
enter 6 sorted elements
enter the element to search:
element found at index 3
```

Q4. Searching

— Not Submitted —

Q5. Array

CODE

```
#include<stdio.h>
int main()
{
    int n,key;
    printf("enter size of array:");
    scanf("%d",&n);
    int arr[n];
    printf("enter %d sorted element:",n);
    for(int i=0;i<n;i++)
        scanf("%d",&arr[i]);
    printf("enter element to search:");
    scanf("%d",&key);
    int low=0,high=n-1;
    while (low<=high)
    {
        int mid=(low+high)/2;
        if (arr[mid]== key){
            printf("element found at index %d\n",mid);
            return 0;
        }
        else if (arr[mid]<key){
            low=mid+1;
        }
        else{
            high=mid-1;
        }
    }
    printf("element not found\n");
    return 0;
}
```

INPUT 6
10 20 30 40 50 60
50

OUTPUT enter size of array:enter 6 sorted element:enter element to search:element found at index 4

Q6. Merging

— Not Submitted —

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Q7. Largest and Smallest element

CODE

```
#include<stdio.h>
int main()
{
    int n,i;
    printf("enter the number of element:");
    scanf("%d",&n);
    int arr[n];
    printf("enter the %d element:",n);
    for(i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    int largest=arr[0];
    int smallest=arr[0];
    for(i=1;i<n;i++)
    {
        if(arr[i]>largest)
            largest=arr[i];
        if(arr[i]<smallest)
            smallest=arr[i];
    }
    printf("largest element :%d\n",largest);
    printf("smallest element:%d\n",smallest);
    return 0;
}
```

INPUT 5
8 9 3 1 3

OUTPUT enter the number of element:enter the 5 element:largest element :9
smallest element:1

Q8. Second Largest Element

— Not Submitted —

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— Not Submitted —

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Q9. Occurrence of an elemenet

CODE

```
#include<stdio.h>
int main()
{
    int n,x;
    printf("enter size of array:");
    scanf("%d",&n);
    int arr[n];
    printf("enter %d elements:",n);
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    printf("enter elements to find:");
    scanf("%d",&x);
    int first=-1,last=-1;
    for(int i=0;i<n;i++){
        if(arr[i]==x)
        {
            first=i;
            break;
        }
    }
    if (first!=-1){
        for(int i=n-1;i>=0;i--){
            if(arr[i]==x)
            {
                last=i;
                break;
            }
        }
    }
    if(first==-1)
    {
        printf("element %d not found in the array.\n",x);
    }
    else{
        printf("first occurrence of %d is at index %d\n",x,first);
        printf("last occurrence of %d is at index %d.\n",x,last);
    }
    return 0;
}
```

INPUT 7
2 5 3 5 7 5 8
5

OUTPUT enter size of array:enter 7 elements:enter elements to find:first occurrence of 5 is at index 1
last occurrence of 5 is at index 5.

Q10. Missing Element

— Not Submitted —